Roll No. .....

TEE–403

B. TECH. (EE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

SIGNALS AND SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you mean by signals ? Discuss energy and power signals. (CO1)

OR

(b) Fundamental periods of periodic signals $x(t), y(t)$ and $z(t)$ are 4 s, 1.25 s, and 2 s, respectively. Check whether the sum of these signals is periodic or not. If its is periodic then find fundamental time period of the resultant signal. (CO1)

2. (a) Determine Energy and Power of the following signals : (CO1)

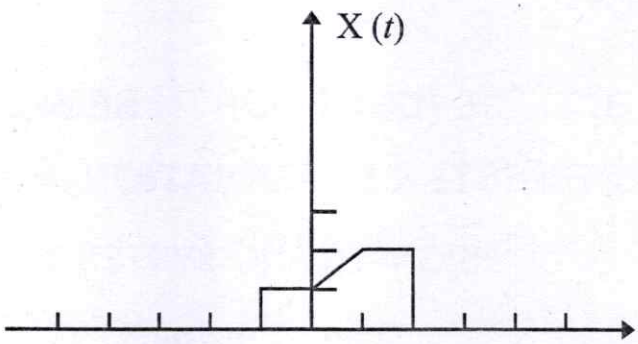
(i) $x(t) = e^{(-2t)} u(t)$

(ii) $x(t) = \mathrm{A}\cos(t)$

P. T. O.

OR

(b) For the signal, X (t) as shown below, find (i) x ($3t$) (ii) x ($2t + 3$) : (CO1)



3. (a) Consider a continuous-time system with input x (t) and output $y(t)$ related by $y(t) = x(\sin(t))$: (CO2)

(i) Is this system causal ?

(ii) Is this system linear ?

OR

(b) If convolution two signal $x(t)$ and $h(t)$ is $y(t)$, then show that convolution of $x(t)$ and $h(t - t_0)$ will be equal to $y(t - t_0)$. (CO2)

4. (a) Find Fourier series expansion of the signal given as : (CO3)

$$x(t) = \frac{10}{2\pi} t; 0 < t < \mathrm{T}$$

OR

(b) Find the convolution of two continuous time signals given as $x(t) = e^{-t^2}$ and $g(t) = 3t^2$. (CO3)

5. (a) Apply the fundamental concept of convolution integral and show that the convolution of odd and even function is odd whereas convolution of two even signals is even. (CO3/CO2)

OR

(b) Discuss the following systems : (CO3/CO2)

(i) Linear systems

(ii) Causal systems

(iii) Time-invariant systems

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Roll No. .....

TEE–402

B. TECH. (EE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

INTRODUCTION TO ELECTRICAL ENERGY SOURCES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) How are stationary power plants classified ? (CO1)

OR

(b) What are the reasons of power crises in India ? (CO1)

2. (a) Draw a typical layout of a coal-based stream plant. (CO2)

OR

(b) Briefly discuss the functions of the following equipments in a steam station : (CO2)
(i) Reheater
(ii) Economiser

3. (a) What considerations must be kept in view in siting thermal plants ? (CO2)

OR

(b) A stream power plant has coal consumption of 165 ton/hr. Calorific value of coal is 3500 kcal/kg. If the power generation is 250 MW, find overall efficiency of the plant. Use relation 1 kcal = 4.18 kJ. (CO2)

P. T. O.

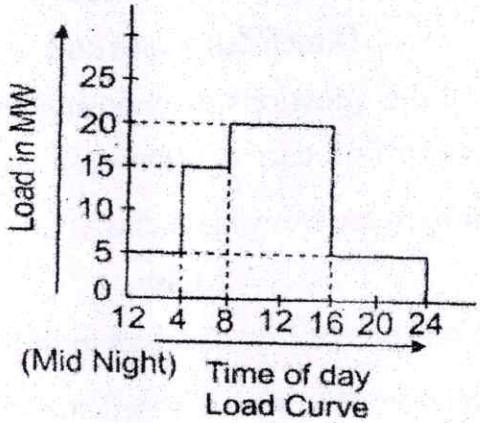
4. (a) What is tariff ? Explain the various types of tariffs. (CO2, CO3)

OR

(b) The yearly load duration of a power plant is a straight line. The maximum load is 30 MW and the minimum load is 20 MW. The capacity of the plant is 35 MW. (CO2, 3)

Find (i) plant capacity factor, (ii) load factor and (iii) utilization factor.

5. (a) Determine the load factor from the following load curve : (CO3)



OR

(b) Incremental fuel cost in rupees per megawatt hour for two units in a plant are given by the following two equations : (CO3)

$$0.1\ P_1+20 \text{ and } 0.12\ P_2+16$$

The minimum and maximum load on each unit is to be 20 MW and 125 MW. Determine incremental fuel cost and allocation of load between units for minimum cost when load is 100 MW.

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Roll No. .....

TMA–401

B. TECH. (ELECTRICAL ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

MATHEMATICS—III

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) An urn contains 10 white, 15 red and 25 black marbles. Determine the probability of drawing (a) Either a white or a red marble. (b) Either a white or a red or a black marble. (CO1)

OR

(b) If 12 persons are seated at a round table, what is the probability that two particular persons sit together ? If 6 boys and 6 girls sit in a row randomly, find the probability that all the 6 girls sit together. (CO1)

P. T. O.

2. (a) In a competitive examination 30 candidates are to be selected. In all 600 candidates appear in written test, and 100 will be called for interview. What is the probability that a person will be called for interview and also find the probability of a person getting selected, if the has been called for interview. (CO1)

OR

(b) The mean and variance of a binomial variate X with parameters n and p are 16 and 8. (CO1)

Find the following :

(i) P (X = 0)

(ii) P (X = 1)

(iii) P (X ≥ 2)

3. (a) The diameter of an water pipe, say X, is assumed to be a continuous random variable with probability density function $f(x) = 6x(1-x), 0 \le x \le 1$. (CO2)

(i) Check validity of $f(x)$ to be p. d. f.

(ii) Determine a number b such that P (X < b) = P (X > b).

OR

(b) The random variable X has a Poisson distribution. If P (X = 1) = 0.01487, P (X = 2) = 0.04461. then find P (X = 3). (CO2)

4. (a) Let X be a continuous random variable with probability density function : (CO2)

$$f(x)=\begin{cases} kx & ; \quad 0 \le x \le 1 \\ k & ; \quad 1 \le x \le 2 \\ -kx+3k & ; \quad 2 \le x \le 3 \\ 0 & ; \quad \text{elsewhere} \end{cases}$$

(i) Determine the value of k.

(ii) Determine mean and variance of X.

OR

(b) The number of airplane landings at an airport in a minute (X) and their probabilities are given by : (CO2)

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X.

5. (a) If X is a random variable for probability mass function

$$p(x)=\begin{cases}\dfrac{x}{15}, & x=1,2,3,\ldots\ldots\ldots\ldots \\ 0, & \text{elsewhere}\end{cases}.$$

find (i) P (X = 1 or 2) and (ii) $P\left(\dfrac{\frac{1}{2}<x<\frac{5}{2}}{X>1}\right)$. (CO1)

OR

(b) Three machines A, B, C produce respectively 60%, 30%, 10% of the total number of items of the factory. The percentages of defective output of this machine are respectively 2%, 3%, 4%. An item is selected at random from the product and is found defective. Find the probability that item was produced by machine C. (CO1)

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Roll No. .....

VAC–401

B. TECH. (EE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

ELECTRICAL SAFETY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the effects of electricity on the human body. (CO1)

OR

(b) What are the primary hazards associated with an arc blast ? (CO1)

2. (a) Discuss the safety standards of electricity in detail. Which one is associated with wiring and earthing ? (CO2)

OR

(b) Why are circuit breakers important ? Explain the difference between a circuit breaker and a relay. (CO2)

3. (a) What are the different Electrical hazards in the context to electrical safety ? Explain any *two* in detail. (CO1, CO2)

OR

(b) Discuss the effect of different current levels on the human body.

(CO1, CO2)

P. T. O.

4. (a) Explain : (CO1, CO2)

(i) CPR

(ii) Electrocution

(iii) Electric shock

OR

(b) List the general safety precautions to be followed to avoid any electrical hazard. (CO1, CO2)

5. (a) Explain : (CO1, CO2)

(i) Electrostatics

(ii) Electromagnetism

(iii) Fuse

OR

(b) What is insulation breakdown ? Give you point of view regarding electrical safety at the construction side. (CO1, CO2)